TRUONG TAN VANG

Software Engineer

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OBJECTIVE

Passionate software engineer with 7+ years of expertise in Android development and automotive systems. Proven track record of leading technical teams, architecting complex solutions, and delivering high-quality software for automotive OEMs, including LG Electronics, GM, Nissan, Renault, BMW, and Land Rover. Specialized in Android Automotive OS, embedded systems, and modern software architecture.

EDUCATION

2015 - 2019

HCMC University of Technology and Education
Bachelor of Engineering
Best Graduation Project Award

WORK EXPERIENCE

8.2025 - Present

LG ELECTRONICS
Coding Expert & Funtional Architect
Project: HKMC– Hyundai, GM

1. GVM SystemUI [Maintenance & Enhancement]
This is a module that inherits the source code of SystemUI AOSP. LGe then modified the features to suit the project.
My notable contribution:
• Rebuild the layout to support multiple themes, vehicle models, and screen sizes.
Create a new QuickControl: a floating panel created using WindowManager to provide quick system settings without interrupting the currently displayed content.
• Modified PowerManagerService: Updated the service to provide necessary data to SystemUI for implementing Minimal Mode and Burn-in protection popups.
• Updated com.android.internal.policy: Implemented changes to block all application content from rendering beneath the Navigation Bar.
• Enhanced Visual Effects: Added ro.surface_flinger.supports_background_blur=1 for PRODUCT_SYSTEM_PROPERTIES to enable the background blur effect when displaying the Quick Control Bar.
• Unit Testing: Utilized JUnit 4 as the primary test runner and Mockito-Kotlin for mocking dependencies in unit tests.

2. Improving Project Developer Environment
• Configure ADB setup directly into Android containers with a mountable system and system_ext partition

• Set up Python scripts to sync source code from local machines to the build server, with integrated automated download and installation.
• Configure Gradle build for SystemUI to enhance UI development and streamline the build and testing workflow.

3. Technical support for modules: Manual HMI, QIBLA HMI, Settings HMI, Vehicle Information, LgeUiLib, Talkback service, Keyboards

Technologies: Linux OS, Android Container, Java, Kotlin, LXC

9.2024 - 8.2025

LG ELECTRONICS
Functional Owner (Technical Leader)
Project: HKMC– Hyundai, BMW

1. WebOS Projection Android BMW [New Development]
A dual-layer framework bridging Android (container) and WebOS (container) via an LCP Bus
• HMI Layer: A Jetpack Compose-based Android Launcher rendered transparently over WebOS. It leverages AIDL Binder for service communication, supports System UID privileges, and exposes a Public API for third-party integration.
Service Layer:
• Projection Service: Acts as a central broker using Proxy/Manager patterns with Coroutine-based auto-reconnection.
• System Service: A JNI Gateway bridging Android to the WebOS native LCP bus.
• Data & IPC: Standardized on AIDL across all layers with JSON polymorphic deserialization for flexible message handling.

2. Technical support for modules: ECall HMI/Service, Manual HMI, QIBLA HMI

Technologies: QNX Hypervisor, Android VM, SOMEIP, Java, Kotlin

7.2023 - 9.2024

LG ELECTRONICS
Functional Owner (Technical Leader)
Project: AIVI – Nissan & Renault (Automotive)

1. HVAC, Seat Services [Maintenance]
These are plug-in services (running in the same process as the car service) that retrieve/set information from the configuration (both live configuration and persistence configuration).
They also contain the logic to provide data for the HVAC/SEAT HMI to function correctly.

2. FoD Service [New Development]
Feature on Demand — handling applicability checks, execution validation, and state persistence for vehicle software features delivered via OCM (Off-board Communication Manager).
My notable contribution:
• State-machine driven execution pipeline — The service manages FOD requests through a dedicated state machine. Ensuring ordered, resumable processing of the FOD lifecycle events even across reboots (state is persisted to mnt/product/persist-alliance/fod/fod_state).
• Configure SEPolicy: Dedicated nissan_fod_file label; rw for alliancecarservice_app only.

3. Technical support for modules: Unit Service, User Profile Service
4. ASPCE support for members: Review all ASPICE documents (SRS, SAD, SDD, TC) for the team's modules

Technologies: Android Automotive OS, Binder, AIDL, IPC, Java

6.2021 - 7.2023

LG ELECTRONICS
Embedded Software Developer
Project: PIVI – Land Rover (Automotive)

1. Online Media - HMI Layer [Maintenance & Enhancement]
An application in the car that allows users to listen to online media content like Spotify, TuneIn, Deezer, Kuwo, Alexa, and Tmall Genie,...
My notable contribution:
• Improved app performance through caching implementation
• Enhanced UX with dynamic background colors based on album artwork

2. Cluster Service – Service Layer [Maintenance & Enhancement]
A service for the exchange of information between the Cluster device and other services of our system. This service provides data about map, intersection, media, phone to display on Cluster device. Data transfer protocols include: SOME/IP, CAN, APIX

Technologies: QNX OS, QT Framework, IPC, C++

6.2019 - 6.2021

DOOSAN Viet Nam
R&D Engineer

1. Mini-Car Inspect inside piping: [New Development] C/C++ (Raspberry) ; C# (Xamarin)
Using a smartphone to control a mini-car via Wi-Fi, the mini-car, equipped with a camera (using a Raspberry Pi, a camera module, a servo, and a DC motor), will go inside the piping and stream video to the smartphone (Xamarin C#).

2. Temperature and humidity monitoring system: [New Development] C/C++(Arduino); C# (WPF)
Read sensor data and transfer it to the computer (via the Ethernet module).
Save data and show the data by line chart (C# WPF).
Built a Xamarin mobile app for Wi-Fi control

Technologies: Xamarin, C#, Pi, WPF

CERTIFICATIONS

6.2025	**Coding Expert at LG Global**
10.2025	**Functional Archiect at LG Global**
12.2024	**AI Coder certification**
6.2022	**Software Programing Competency Test**

HONORS & AWARDS

2018	**Best Graduation Project Award**
2025	**Best Member Award 2025**

SKILLS

Core Competencies	Proactive in learning and adapting to new information rapidly. Highly proficient in business logic
Programing langue	Java/Kotlin/C++ to develop apps, service. Python to create helper scripts
OS/Container	Android OS, Android Container, QNX Hypervisor, Linux OS
Build Tools	Gralde / Soong Build system
Management	Agile/ASPICE
AI Tools	Github Copilot
Git	Github, Gitlab
Algorithm	Strong knowledge about Data structures and Algorithms
Other skills	SELinux, Customize Android SDK, SWUT

PERSONAL PROJECTS

2018 - 2019

AI Massage Robot (Graduation Project)

First Prize – 2018 Student Scientific Research Award

https://vnexpress.net/sinh-vien-tp-hcm-dau-tu-20-trieu-dong-che-tao-giuong-massage-3857654.html